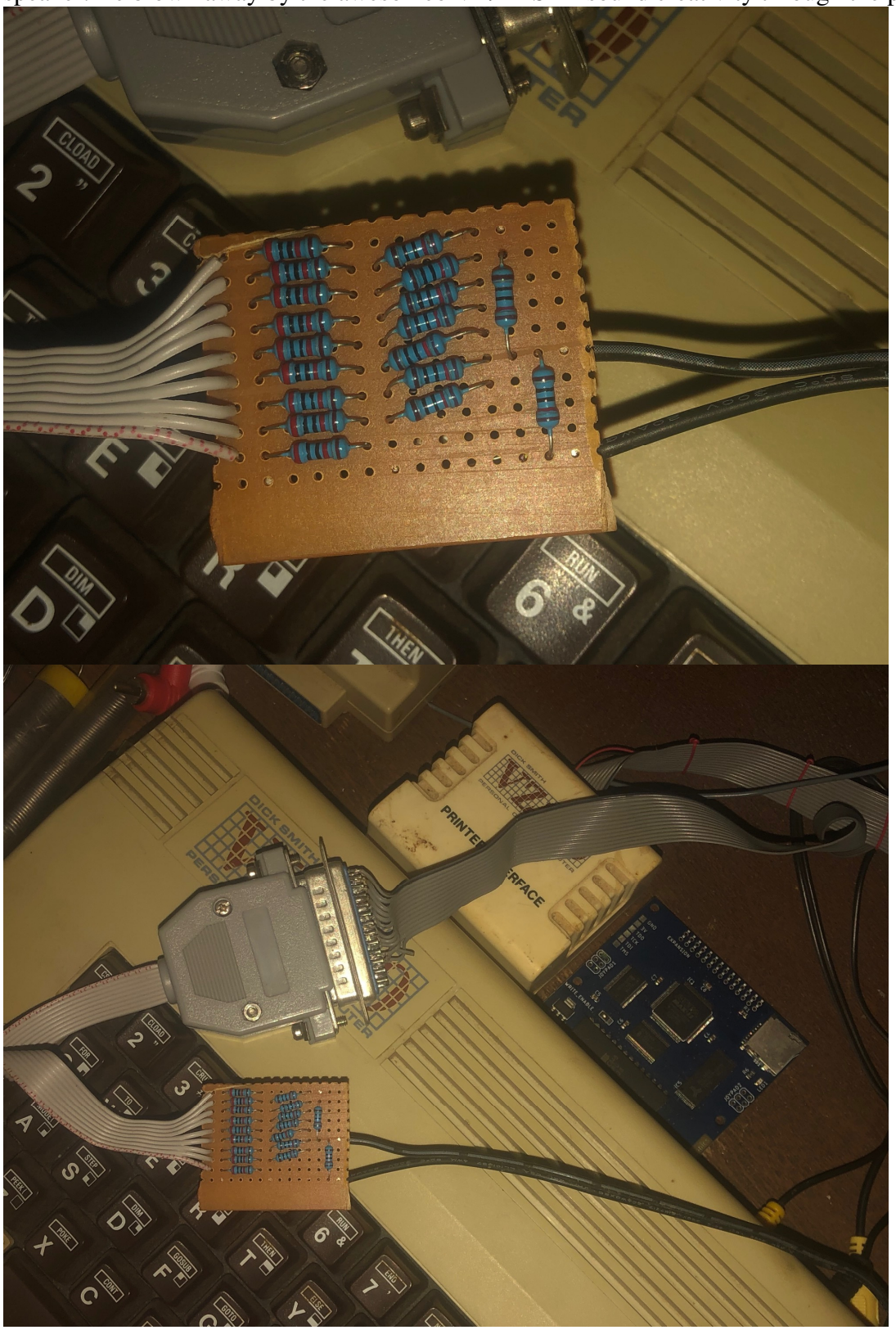


Quick write up to for music and audio generation through the printer interface to a connected resistor ladder or IC DAC. Connect to amplified speakers, home stereo unit, or easily add a small op-amp with speaker. Be blown away by the awesomeo VZ/LASER sound creativity through the printer interface.



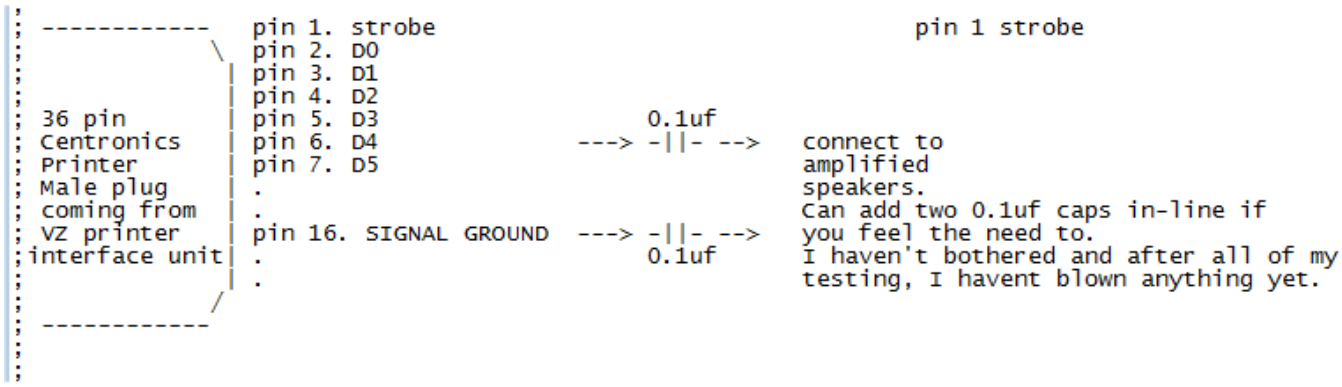
VZ200 printer interface with connected 8 bit DAC with SD Loader.

Experimental Printer Port option – 1-bit music

Just connect up your VZ/Laser PI20 or PI30 printer interface to your VZ/Laser.
Then with a centronics female socket, connect D4 and GND to an amplified speaker via two 0.1uf caps to protect everything. Im not using the caps in my DAC, but I am telling you to use them,
D4 being output data bit 5, being pin 6 on the Centronics plug.

Chose data line D4/bit5 since bit5 is the same data line on the $6800 latch for the internal speaker. We may as well make the printer port use the same, since it just makes outputing data just that little bit easier. We could choose any of the 8 data lines, but why make things harder than they need to be.
LD ($6800), A --> OUT (14), A

The Printer Interface sits on the first 16 ports, but port 14 is the official printer interface port, so we use it. The existing music written for bits 1 and 5 for the latch for the internal speaker can now just be sent straight to Port 14 using OUT(14), A. Or, LD C, 14 / OUT (C), A
Register A contains the 1bit music data.

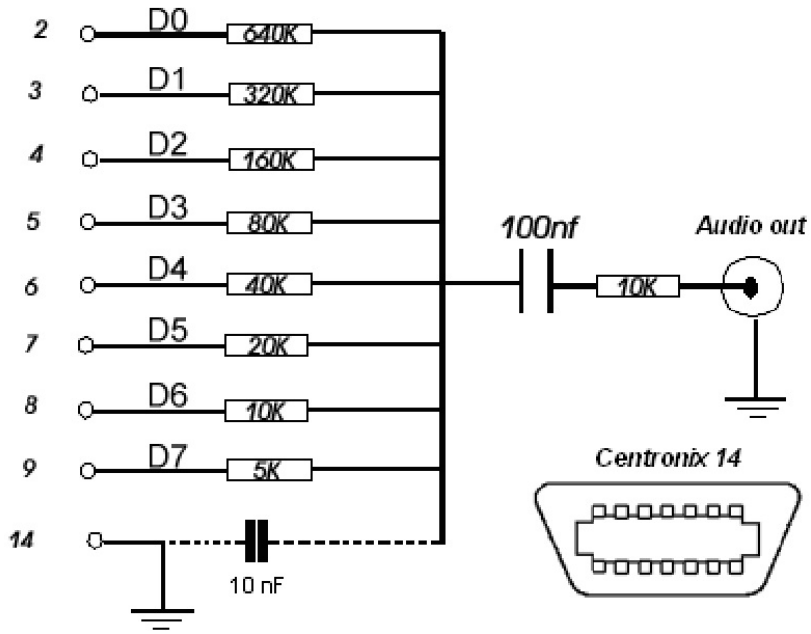


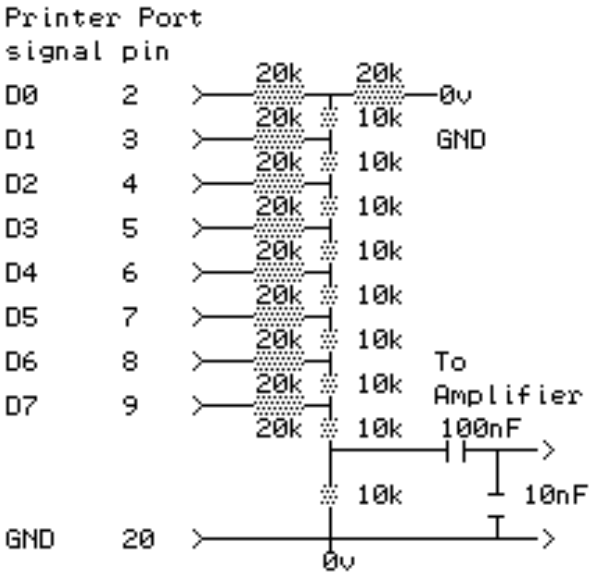
I have modified a number of music programs that will use the printer interface to play music (as well as using the normal internal speaker and even using the cassette port). Can be found in the 1-bit music directory over in: https://github.com/bushy555/VZ200/tree/master/hardware/sound/

A few of the most nicest sounding pieces of music are:
- Lets Go by Shiru.
- Night Dancer by Mister Beep.
- Beeper Drive by Dmilty_Milk.

Experimental Printer Port option – 8-bit music on a Resistor ladder

There are numerous methods in making a simple resistor DAC ladder circuit.
Can be using resistor of the same value, (natural slope) , or, of varying expotential difference in values being a logarithmic slope sound output. Tone, timbre, base and pitch sound qualities will all be slightly effected when compared between the two DAC's.





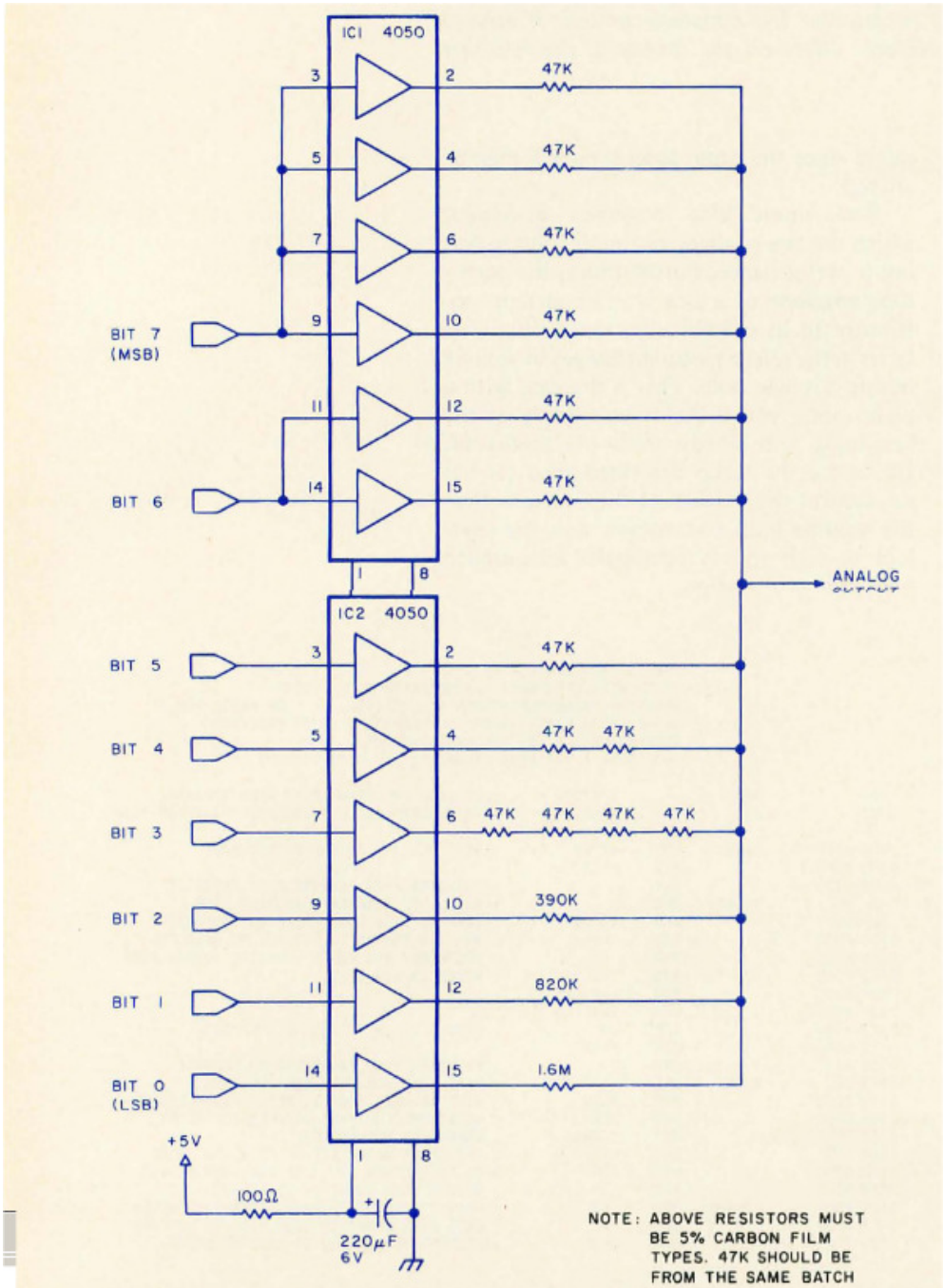


Figure 3: An 8 bit digital to analog converter (DAC). This circuit accepts an 8 bit binary number from the output port and generates a corresponding DC voltage as its output. The output voltage from this circuit is equal to ((I/255)-x5) V, where I is the decimal equivalent of the 8 bit input which can take on any value from 0 to 225.

A number of Amiga .MOD music module files, that are under 30kb in size, have been modified to play through an 8-bit Printer Interface DAC. Some noise is to be expected from both heavily cut down original 8-bit quality samples to fit into our limited space, and, well… this is completely outside what the original Vtech engineers had in mind for the mightly little VZ/Laser , we are venturing way beyond their documented designed SOUND 31,1;27,1;28,2 !!
MOD Files, with their assembly sources for further experimenting and expansion, can be found : https://github.com/bushy555/VZ200/tree/master/hardware/sound